

GoalOS Mission OS

The Proof OS for Autonomous AI Work

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v12 – Commercial Proof-to-Action Edition

Canonical line

Set the objective. GoalOS runs until proof is done.

AI creates output. GoalOS creates proof. No proof, no settlement. No eval, no propagation. No rollback, no release.

1 Executive thesis

GoalOS Mission OS is the commercial front end of the Proof OS for AI Work. It turns a high-stakes objective into a governed decision state: evidence, verification, risk, action, rollback conditions, Chronicle memory, and reusable capability. The product is not another report generator. The product is a mission-native operating system that runs until proof is done.

2 The primitive: Governed Decision State

A report is static. A Governed Decision State is live, inspectable, and operational. It contains the mission objective, decision to support, claims matrix, source provenance, contradiction register, Evidence Docket, verifier report, risk ledger, action graph, rollback conditions, Chronicle entry, capability package, and next mission recommendation.

3 Mission value

GoalOS optimizes mission value rather than report length:

$$V_{mission} = \frac{I_{decision} P_{integrity} A_{actionability} R_{reuse}}{C_{cost} + R_{risk} + L_{latency} + D_{proof\ debt}}. \quad (1)$$

The highest-value output is not the longest report. It is the shortest path from uncertainty to justified action.

4 AI work equation

$$W_{AI} = O \times P \times V \times S \times R, \quad (2)$$

$$\text{where } O = \text{output}, \quad P = \text{proof}, \quad V = \text{validation}, \quad (3)$$

$$S = \text{settlement or state transition}, \quad R = \text{reuse}. \quad (4)$$

If proof is zero, institutional work is zero:

$$P = 0 \quad \Rightarrow \quad W_{trusted} = 0. \quad (5)$$

5 Mission lifecycle

$$\text{Objective} \rightarrow \text{Mission Contract} \rightarrow \text{Planning} \quad (6)$$

$$\rightarrow \text{Autonomous Work} \rightarrow \text{Verification} \quad (7)$$

$$\rightarrow \text{Evidence Docket} \rightarrow \text{Decision State} \quad (8)$$

$$\rightarrow \text{Action Graph} \rightarrow \text{Chronicle}. \quad (9)$$

6 Acceptance law

A mission artifact is accepted if and only if all mandatory gates clear:

$$Accept(m) = 1 \iff ProofBundle(m) \wedge ValidatorClear(m) \quad (10)$$

$$\wedge SourceProvenance(m) \wedge ContradictionsLogged(m) \quad (11)$$

$$\wedge RiskLedger(m) \wedge ActionGraph(m) \quad (12)$$

$$\wedge ClaimBoundaryPass(m) \wedge ReviewReady(m). \quad (13)$$

Otherwise, $Accept(m) = 0$.

7 Until-done autonomy

Autonomy is an artifact-generation loop, not a governance bypass. The run state is:

$$\Sigma_t = (M_t, A_t, Q_t, B_t, D_t), \quad (14)$$

where M_t is mission state, A_t generated artifacts, Q_t quality gates, B_t boundary checks, and D_t done status. The system advances by repair:

$$\Sigma_{t+1} = Repair(Generate(\Sigma_t), Q_t, B_t). \quad (15)$$

The loop halts only when:

$$DONE = 1 \iff RequiredArtifactsComplete \quad (16)$$

$$\wedge EvidenceComplete \wedge BoundaryPass \quad (17)$$

$$\wedge QAReportPass \wedge HumanReviewReady. \quad (18)$$

8 Verifier Mesh

The Verifier Mesh scores the integrity of the mission package:

$$\mathcal{V}(m) = w_s S_{sources} + w_c C_{claims} + w_x X_{contradictions} \quad (19)$$

$$+ w_r R_{risk} + w_a A_{action} + w_b B_{boundary}. \quad (20)$$

A verifier is useful only when it blocks unsupported propagation. Scores are advisory; gates are mandatory.

9 Evidence Graph moat

Let the Evidence Graph be:

$$EG_t = \{Missions, Claims, Sources, Proofs, Reviews, Risks, Actions, Outcomes, Capabilities\}_{0:t}. \quad (21)$$

The moat is the time-integrated quality of the graph:

$$Moat(t) = \int_0^t \left(Q_{evidence} R_{reuse} X_{review} S_{settlement} \right) d\tau \quad (22)$$

$$- D_{proof\ debt} - K_{copyability}. \quad (23)$$

A demo can be copied. A living evidence graph cannot be retroactively fabricated.

10 Product ladder

Layer	Commercial meaning
Mission Room	One objective becomes one proof-backed decision state.
Mission OS	Recurring mission operations across a team or organization.
Evidence Vault	System of record for missions, proofs, decisions, risks, and capability packages.
Verifier Mesh	Review, challenge, and validate claims before action.
Chronicle	Institutional memory of what proved useful.
AGIALPHA Settlement	Optional proof-settlement layer where value movement remains human-gated.
Readiness	

11 Claim boundary

GoalOS Mission OS does not claim achieved AGI, ASI, superintelligence, guaranteed ROI, legal advice, financial advice, medical advice, production readiness, external audit completion, cybersecurity certification, mainnet deployment, or guaranteed market outcomes. The claim is product-operational: a mission-native system can generate proof-backed decision states and package evidence for human-governed review.

12 Canonical conclusion

Research ends with a report. GoalOS ends with a governed decision state. Insight is not finished until it becomes governed action. Set the objective. GoalOS runs until proof is done.